

SIMFEROPOL, Ukraine

WMO: 339460

Lat: 45.0403N

Lon: 33.9672E

Elev: 181

StdP: 99.17

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.2	-9.3	-15.1	1.0	-11.4	-12.6	1.3	-8.4	15.8	-4.1	14.4	-3.1	6.1	50	0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	33.8	20.5	32.0	20.1	30.2	19.5	22.5	28.8	21.6	28.0	20.9	27.3	6.0	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.8	15.8	24.8	19.8	14.8	24.2	18.9	14.0	23.4	67.1	28.4	63.9	27.9	61.1	27.4	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.6	11.0	9.6	-15.9	36.2	4.0	1.5	-18.7	37.3	-21.1	38.2	-23.3	39.0	-26.2	40.1		
(5)			-16.1	23.8	3.9	0.9	-18.9	24.4	-21.2	25.0	-23.3	25.5	-26.2	26.1		

Monthly Climatic Design Conditions

			Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
			(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.7	0.5	1.4	5.0	10.6	16.0	20.4	23.3	23.2	17.9	11.9	7.1	2.6	(6)
(7)		DBStd	9.12	5.46	5.65	4.51	3.94	3.59	2.99	2.65	3.06	3.58	4.42	4.78	5.33	(7)
(8)		HDD10.0	1121	297	243	164	38	2	0	0	0	1	31	111	234	(8)
(9)		HDD18.3	2854	554	473	413	233	89	13	1	2	49	203	336	487	(9)
(10)		CDD10.0	1749	1	3	10	56	188	312	413	408	238	90	25	6	(10)
(11)		CDD18.3	442	0	0	0	1	17	74	156	152	37	4	0	0	(11)
(12)		CDH23.3	4201	0	0	1	19	172	655	1503	1509	306	34	2	0	(12)
(13)		CDH26.7	1565	0	0	0	2	29	198	610	646	74	6	0	0	(13)
(14)	Wind	WSAvg	4.5	5.0	5.4	5.1	4.8	4.1	4.0	4.1	4.0	4.3	4.4	4.6	4.9	(14)
(15)	Precipitation	PrecAvg	495	42	33	36	32	39	54	50	45	37	39	44	48	(15)
(16)		PrecMax	782	110	86	93	95	138	130	161	291	96	123	111	112	(16)
(17)		PrecMin	309	6	5	2	0	2	3	0	0	0	3	2	3	(17)
(18)		PrecStd	104	27	19	21	22	28	31	39	47	22	27	27	26	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	17.1	20.2	25.9	29.6	33.2	35.8	36.2	31.7	27.0	22.3	17.0	(19)
(20)			MCWB	9.2	10.1	10.8	14.9	18.2	20.2	20.9	20.5	19.0	16.6	14.0	10.8	(20)
(21)		2%	DB	11.8	13.7	17.1	22.7	27.1	31.0	33.7	34.0	28.8	23.8	19.0	13.9	(21)
(22)			MCWB	8.4	9.1	10.0	13.4	17.5	19.5	20.6	20.5	18.1	15.7	12.7	10.0	(22)
(23)		5%	DB	9.5	11.2	14.7	20.2	25.2	28.9	32.0	32.2	26.8	21.1	16.2	11.9	(23)
(24)			MCWB	7.2	8.0	8.8	12.4	16.8	19.1	20.4	20.0	17.3	14.7	11.8	9.3	(24)
(25)		10%	DB	7.7	9.1	12.1	17.9	23.2	27.1	30.1	30.3	24.9	18.7	13.9	9.9	(25)
(26)			MCWB	6.1	6.6	7.8	11.3	16.1	18.5	19.8	19.5	16.8	13.9	10.7	8.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	11.2	12.0	16.0	20.2	22.7	23.6	23.9	21.3	18.3	15.2	12.3	(27)
(28)			MCDWB	13.1	15.2	17.9	23.8	26.0	29.6	29.8	30.4	26.4	23.8	19.3	15.3	(28)
(29)		2%	WB	8.7	9.7	10.5	14.3	18.8	21.2	22.5	22.5	19.7	16.8	13.6	10.7	(29)
(30)			MCDWB	10.9	12.9	15.6	21.2	24.9	27.0	28.6	29.0	25.1	21.1	17.8	13.1	(30)
(31)		5%	WB	7.4	8.3	9.4	13.0	17.7	20.3	21.6	21.6	18.7	15.7	12.3	9.6	(31)
(32)			MCDWB	9.3	11.0	13.7	18.8	23.2	26.0	28.0	28.3	24.1	19.6	15.6	11.6	(32)
(33)		10%	WB	6.0	6.7	8.2	11.9	16.7	19.5	20.9	20.7	17.7	14.4	11.0	8.1	(33)
(34)			MCDWB	7.4	8.7	11.6	17.0	22.0	25.0	27.5	27.4	22.8	18.1	13.5	9.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.5	7.3	8.9	10.8	11.3	11.1	11.9	12.2	11.0	9.4	7.9	6.4	(35)
(36)			MCDBR	9.2	10.6	13.1	14.2	14.0	13.7	14.3	14.6	14.0	12.7	11.4	8.8	(36)
(37)			MCWBR	6.5	7.0	7.6	7.5	6.8	5.6	5.3	5.5	5.9	6.4	6.7	6.2	(37)
(38)		5% WB	MCDBR	8.6	9.5	11.6	12.7	12.7	11.6	12.4	12.8	11.8	11.3	10.0	8.0	(38)
(39)			MCWBR	6.5	6.6	7.2	7.2	6.7	5.5	5.3	5.7	5.9	6.4	6.5	6.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.335	0.360	0.387	0.426	0.404	0.405	0.424	0.425	0.399	0.366	0.347	0.333	(40)	
(41)		taud	2.355	2.286	2.234	2.148	2.299	2.329	2.290	2.273	2.319	2.385	2.379	2.362	(41)	
(42)		Ebn at Noon	763	809	837	833	867	864	842	826	817	795	741	722	(42)	
(43)		Edh at Noon	78	100	121	143	127	124	128	125	110	90	76	71	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.14	1.99	3.30	4.99	6.28	6.72	6.79	5.96	4.36	2.70	1.47	0.99	(44)	
(45)		RadStd	0.15	0.24	0.30	0.45	0.45	0.42	0.31	0.34	0.36	0.26	0.16	0.13	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.62	N/A	N/A	N/A	+0.54	+0.56	-102	-138	+116	+64
(47)	Regional (5 neighbors)	+0.67	N/A	N/A	+0.86	N/A	N/A	-119	-167	+141	+87

Nomenclature: See separate page